

Layering and phase transition of liquid aluminum confined by different substrates



Yunrui Duan, Jie Li, Tao Li, Zhichao Wang, Hui Li*

Key Laboratory for Liquid-Solid Structural Evolution and Processing of Materials, Ministry of Education, Shandong University, Jinan 250061, People's Republic of China

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ABSTRACT

Molecular dynamics simulation has been performed to study the layering in the nano-confined liquid aluminum, a magic phenomenon characterized by oscillatory density profile across the molecular thin films. Carbon and copper walls are served as the confinement substrates, which can bring about discrepancy in the layering behavior, coordination number and diffusion coefficient. We highlight the abnormal changes of potential energy and volume when the new layer forms, accompanying the distinctive changes of the average coordination number and diffusion coefficient, which is a typical character of phase transition. The role of pressure and temperature on the layering is also reported. Specifically, the pressure-confinement and temperature-confinement diagrams are plotted to further analyze the layering transition.

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1. Introduction

The properties of material under confinement [1–10], which are distinct from those of the bulk, have received extraordinary research efforts due to their fundamental significance in both science and engineering. Examples include phase transitions, where the confinement can bring about the temperature shift of the freezing, melting and glass phase transition points [11–16]; self-assembly, where the confinement have its unprecedented potential in the area of device fabrications, thus precisely regulating internal structures of the target materials at the mesoscopic, nanoscopic and molecular levels [17–20]; and chemical reactions, where the confinement within carbon nanotubes enables the control of catalyst activity, morphology and stability [1]. Owing to a myriad of practical applications, investigations on properties of materials within confined space become remarkably significant.

The layering, a magic phenomenon under confinement, has been reported in terms of its non-uniform system density across the molecular thin films by Persson et al. [21]. Subsequent researches support the picture that the layering plays a key role in a multitude of fields. For instance, the layering of water molecules surrounding metal nanoparticle can lead to a greater thermal conductivity, thus increasing the heat transfer efficiency of the fluid [22]. The layering of alternating anions and cations in ionic liquid [23–26] is highly correlated with lubrication and catalysis.

Furthermore, the layering of hydrogen isotopes in the inner surface of carbon nanotubes can be applied for the separation of hydrogen isotopes [27]. Even in the dentistry, the layering techniques can be employed to achieve aesthetic purpose [28]. Considering the general significance and wide application, it is surprising that some details of the layering are poorly understood.

Another interesting problem is the mechanism of the layering transition, about which a considerable number of substances were reported. A first order phase transition [29] occurred involving the transformation between a bi-layer liquid and a tri-layer heterogeneous fluid in the confined water. Besides water [30–36], Saslow et al. [37] studied the Helium films with respect to its mono-layer to bi-layer transition using a Maxwell construction. In addition, layering transitions in the confined carbon [38], silicon [39], and silicon carbide [40] liquid films were also clarified with a noticeable structure change. Although remarkable advances have been made, the proofs of the layering transition are still limited especially in some liquid metals. Aluminum (Al) and its alloys are perceived as the most versatile and acceptable engineering materials which are widely used in aerospace components, automobile components and electrical appliances [41,42]. In present work, we confine the liquid Al between two parallel walls and observe its layering transition controlled by slit sizes (SS) and pressure. This simulation results can give insight into the properties of the liquid metal under confinement.

* Corresponding author.

E-mail address: lihuilmy@sdu.edu.cn (H. Li).

2. Models and methods

MD simulations [43–45] are performed using the software package LAMMPS to study the structure of the liquid Al confined by carbon and copper walls. The interactions between the carbon walls and the liquid Al, which is initially arranged into $15 \times 15 \times 2$ face-centered cubic (FCC) lattice, are represented by 6–12 Lennard-Jones (LJ) potentials with the following parameters: $\epsilon_{\text{Al}^- \text{wall}} = 0.038$ eV, $\sigma_{\text{Al}^- \text{wall}} = 2.96$ Å [46]. It is well known that the simulation results are highly dependent on the adopted potential functions and the previous studies mainly adopt Lennard-Jones potential as the carbon walls. In this study, the copper walls, whose interaction with the liquid Al is embedded-atom method (EAM), are also employed, where 2403 Al atoms are used. The pair interaction between metal particles is modeled with the EAM which is particularly appropriate for metallic systems.

Simulations are carried out using the NPT ensemble. The Nose-Hoover thermostat and barostat are adopted to control temperature and pressure, respectively, and the velocity-verlet algorithm is applied to integrate the equations of motion with a time step of 1 fs. The constant pressure parallel to walls (P_{xy}) is exerted to analyze the effect of pressure on the layering. Periodic boundary conditions are applied along the x and y directions. Along the z direction, the confined wall is imposed with the SS varying from 11 Å to 18 Å.

The liquid Al is equilibrated to obtain a relatively homogeneous liquid at high temperature of 2000 K. Then it is cooled to the target temperature (1300 K or 1200 K) at the cooling rate of 1 K/ps. Subsequently, the metal Al is relaxed at the target temperature for 2000 ps, which is adequate for the system to reach its equilibrium.

3. Results and discussions

Fig. 1(a) and (b) shows the density distribution function, which is calculated by counting the atom number in a thin slice with the thickness of 0.05 Å in the z direction, and the corresponding structure snapshots for the liquid Al confined by LJ interaction carbon

walls at 1200 K and copper walls at 1300 K respectively. Each density peak corresponds to an atom layer in the confined liquid as shown by dashed lines. The confined atoms aggregate at three or four density peaks, displaying an outstanding discontinuous distribution, contrary to the conventional wisdom that atoms can scatter uniformly in a free space. From Fig. 1(b), we can see that the density peaks adjacent to the confined walls are higher due to the stronger confinement effect, indicating that the confinement is essential for the layering. The density value of the liquid Al confined by copper walls (Fig. 1(b)) is larger than that confined by the carbon walls, suggesting the confinement effect of copper walls is stronger.

The alteration of SS can lead to the change of layer numbers as displayed in Fig. 2. For the liquid Al confined by carbon walls (Fig. 2(a)), the system keeps tri-layer structure at the SS range from 11 to 12.5 Å, which is named as the tri-layer zone. When the SS increases, the system starts to transform from three to four layers. Upon the SS increasing to 13 Å, the system totally transforms to four layers. Therefore, the SS interval from 12.5 to 13 Å is defined as the transition zone (TZ). The rest may be deduced by analogy, such as the transformation from four to five layers. However, when the liquid Al is confined by copper walls (Fig. 2(b)), the system transforms from four to five layers directly at the SS of 12.5 Å. As a result, the TZ disappears. Besides, the layer number is different under different slits with the same SS. For example, when the SS is 14 Å, the system shows tetra-layer and penta-layer structures for carbon and copper substrates respectively because the confinement extent is different. The system potential energy and volume as a function of SS are also displayed in Fig. 2. It can be found from Fig. 2(a) that at 1200 K, when the SS increases from 11 to 12.3 Å, the potential energy shows no big change. However, a sudden drop of the system potential takes place at the TZ, indicating the occurrence of the layering transition. A steady period of the potential energy at the tetra-layer zone is subsequently followed by another drop, where another layering transition also takes place. Fig. 2(a) also shows the system volume as a function of SS. It is instinctively understandable that the system volume would increase upon the

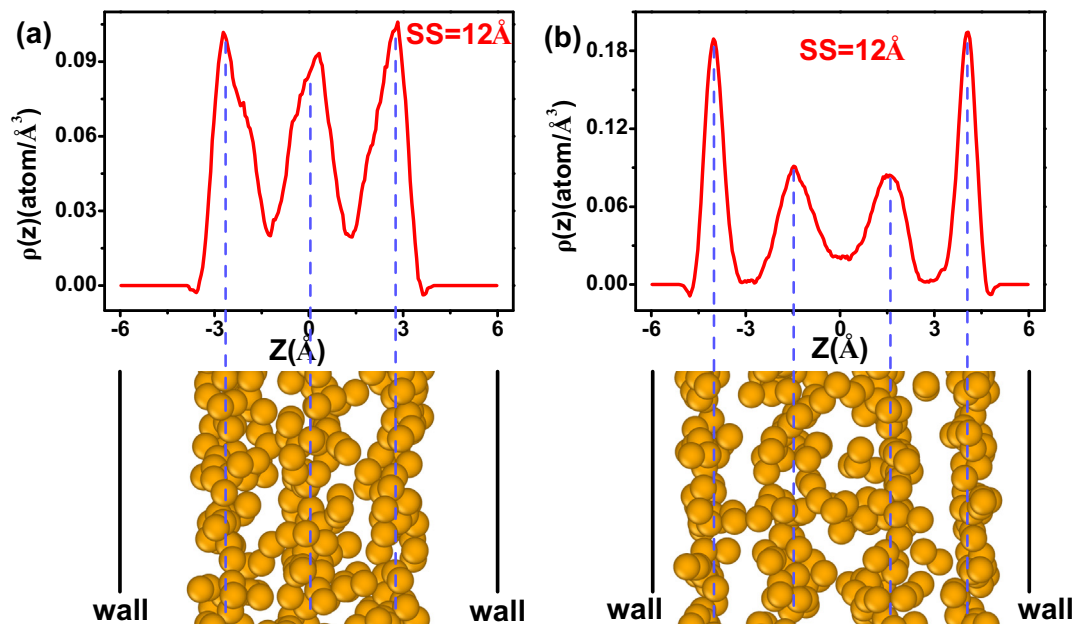


Fig. 1. Density distribution functions (ρ) and snapshots along the z direction of the Al atoms (a) confined by carbon walls when $T = 1200$ K, $P_{xy} = 0$ bars (P_{xy} refers to the lateral pressure along x and y direction), $SS = 12$ Å; (b) confined by copper walls when $T = 1300$ K, $P_{xy} = 0$ bars, $SS = 12$ Å. The confining walls are located at $z = -6$ Å and $z = 6$ Å as displayed by the solid black lines. The peaks in density profiles correspond to the layer locations in the snapshots, which are marked by dashed line. The system exhibits tri-layer and tetra-layer structure in (a) and (b) respectively.

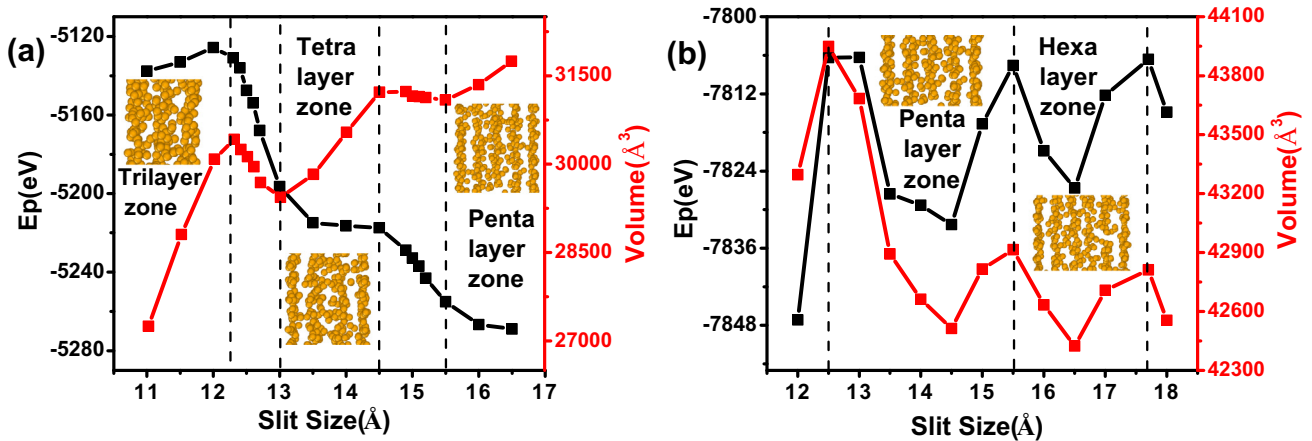


Fig. 2. The system potential energy (E_p) and the volume as a function of the slit size without lateral pressure applied (a) At 1200 K when the confining wall is carbon; (b) At 1300 K when the confining wall is copper. The dashed lines suggest the place where the layer number changes. And TZ in (a) refers to transition zone.

SS increasing. However, it shows abnormal decrease at the TZ, again indicating the phase transition. When the Al atoms are confined by copper walls at 1300 K, both the potential energy and volume (Fig. 2(b)) exhibit an oscillation with the changing SS. Interestingly, the system potential energy and volume reach maximums at the transition points, which is also an indicator of the phase transition. Therefore, it can be concluded that when the layer number increases, the layer transition happens, accompanying abnormal changes of the system potential energy and the volume, which are characteristic indications of the phase transition. Upon the occurrence of the layering transition, the system potential energy and the volume show a greater change at a small SS than that at the larger SS, suggesting that the small SS has great effect on the layering transition because the confinement effect of the small SS is strong.

To further investigate the structure changes during the transition, Fig. 3 displays the average coordination number of the system which is calculated by counting the atom number within the first shell. For the liquid Al at 1200 K confined by carbon walls (Fig. 3(a)), the coordination number changes slowly in the tri-layer, tetra-layer and penta-layer zones. However, in the TZ, it increases rapidly and nearly linearly with the SS. Therefore, the coordination number of the system displays different features in the TZ. While for the liquid Al at 1300 K confined by copper walls (Fig. 3(b)), the coordination number oscillates with the SS. And the minimums of the coordination number happen to correspond to the transition points. Under the two substrates, the coordination number reveals different changing trends despite both giving the evidence of phase

transition. For Al atoms confined by carbon walls, the coordination number varies rapidly in small SS regions but slowly subsequently, such as the contrast between SS intervals from 12 to 13 Å and 16 to 17 Å. Nevertheless, the coordination number curve varies gently when the SS is small but sharply in the large SS region. The disparate changing trends result from the different confinement effect.

Fig. 4 illustrates the coordination number as well as the average potential energy of each layer for the liquid Al confined by carbon walls during cooling process when the system is penta-layer structure. The nomenclature of each layer is displayed in the inset in Fig. 4(b): the middle layer is defined as the central layer and the layer above is then defined as the first layer according to the coordinate values z and so on. We can see that the coordination number of the system increases upon cooling, forming a denser structure. The coordination numbers of the first and minus first, second and minus second layer are almost the same, both larger than that of the central layer. Fig. 4(a) also reveals that the atoms adjacent to the wall have a larger coordination number than those in the distance due to the stronger confinement effect near the wall. Therefore, in the layering structure, the compact degree of each layer is different. And the layer close to the confined wall tends to form a denser structure with larger coordination number. It can be found in Fig. 4(b) that the energy distribution is symmetrical with respect to the central line of the confined space because of the symmetrical confinement. That is, the average potential energy of the first layer and the minus first layer, second and minus second layer are almost the same. It can be seen that the energy

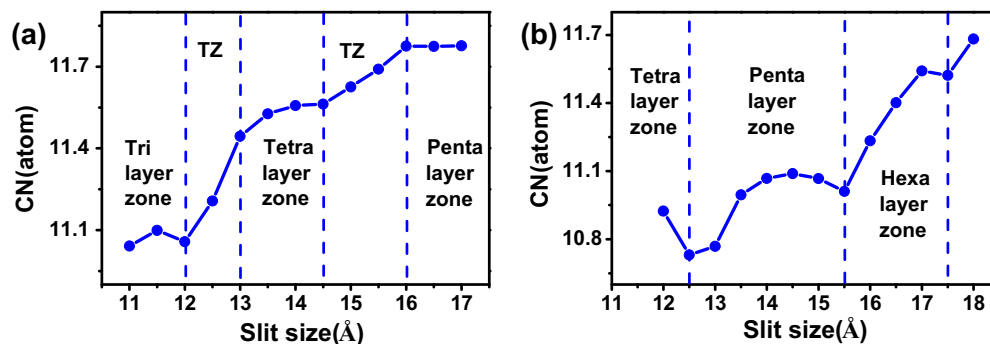


Fig. 3. The average coordination number (CN) as a function of the slit size for the system at (a) $T = 1200$ K, $P_{xy} = 0$ bars when the confining wall is carbon; (b) $T = 1300$ K, $P_{xy} = 0$ bars when the confining wall is copper. The coordination number is calculated by counting the atom number within the first shell and the meaning of the dashed lines and TZ is the same as those in Fig. 2.

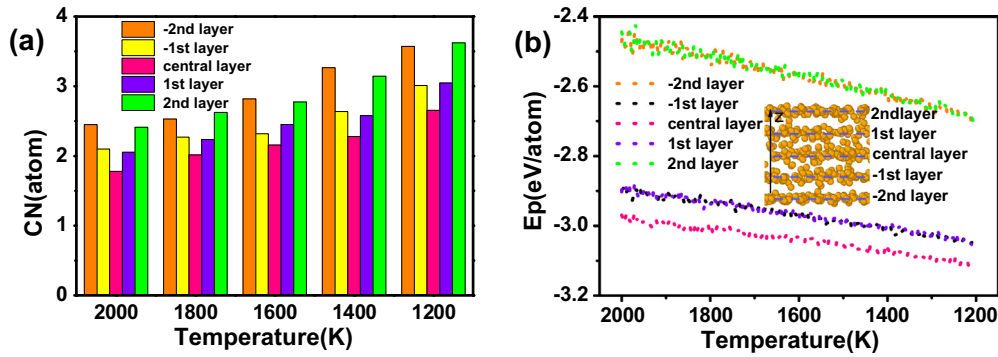


Fig. 4. (a) Average coordination number (CN); (b) average Potential energy (Ep); for Al atoms of each layer during the cooling process when SS = 16 Å, Pxy = 0 bars, where the confining wall is carbon and the system shows penta-layer structure. As what (b) shows, the middle layer is defined as the central layer and the layer above is then defined as the first layer according to the coordinate values z . The rest layer may be deduced by analogy.

difference between the first and second layer is large, displaying a discontinuous distribution. The further the atom layer away from the confined walls, the smaller the energy difference between the adjacent layers, making the system continuous. We can deduce that when the slit size becomes big enough, the layering at the middle of the confined space would become not obvious any more due to the weak confinement effect, which is similar to the bulk system.

In addition to the coordination number, the diffusion coefficient along X axis can also support the phase transition. It is computed by taking derivative of the mean square displacement (MSD) as displayed in Fig. 5. When the Al atoms are confined by the carbon walls at 1200 K (Fig. 5(a)), the diffusivity becomes stronger with the SS increasing because the confinement effect becomes weaker. In the TZ, however, the diffusion coefficient displays abnormal decrease, suggesting its phase transition. However as for the Al atoms confined by the copper walls at 1300 K as shown in Fig. 5 (b), their diffusion coefficient also shows some oscillations upon the increasing SS. And at the transition point, the maximum of the diffusion coefficient emerges, indicating the sudden change of the system state. It is worth noting that the diffusion coefficient in Fig. 5(b) is smaller than that in Fig. 5(a) even if the temperature is 100 K higher, suggesting that the confinement effect of the copper wall is stronger than that of the carbon wall. The strong confinement effect restricts the diffusive property along the x direction.

Next, the effect of pressure and temperature on the layering is studied. Fig. 6(a) shows the density distribution function along the z direction under various pressures which are imposed laterally along both the x and y directions (Pxy) when the SS and the

temperature are fixed. The density distribution function can give a direct description of the layering extent. When the pressure is increased from 0 to 10^4 bar, the density peaks show a slight increase, indicating that the dispersed atoms between two layers decrease, leading to a more apparent layering. When the pressure is increased to 10^5 bar, the density peaks become top and sharp, suggesting that atoms aggregate into several layers. The trough value of the density is absolutely zero, revealing that there are no scattered atoms between two layers. When the pressure is up to 10^6 bar, more obvious layering is observed. Meanwhile, the layer number transforms from four to five. Therefore, the pressure can strengthen the layering and lead to layering transformation when it is high enough. Fig. 6(b) illustrates how temperature affects the layering. Upon temperature reducing, the density peaks become higher and a more obvious layering is found. Comparing with the pressure, the effect of temperature on the layering is slight. Firstly, the positions of density peaks and troughs keep unchanged with the temperature. Secondly, the peak height varies slightly with temperature not like the pressure which makes the peak height increase by orders of magnitude. Thirdly, the temperature inevitably causes layer transition when the pressure and SS is fixed. High temperature is not beneficial for enhancing the layering because the atoms at high temperature have enough kinetic energy to overcome the restriction from the wall to move freely.

Taking all the above discussions into consideration, we plot the layering transition diagram of the liquid Al confined by carbon walls in order to well understand the transition. Fig. 7(a) shows the pressure-SS diagram, revealing that small SS and low pressure are beneficial for decreasing the layer number of the system. At the low pressure zone, the layering transition line changes slightly

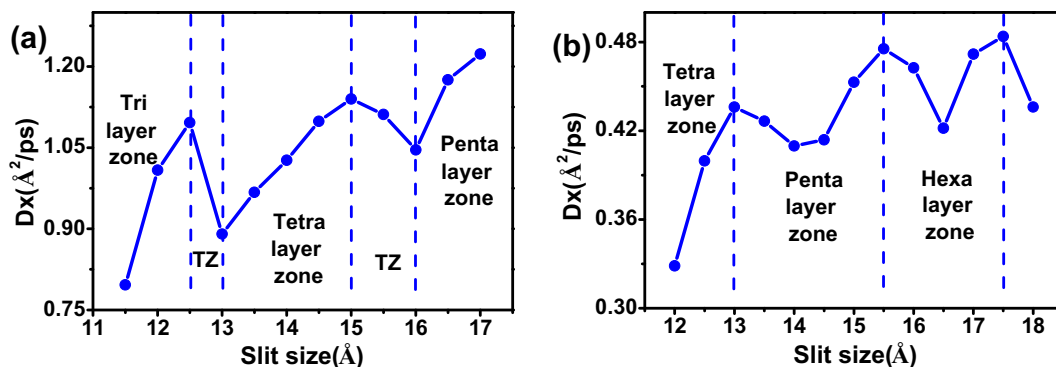


Fig. 5. The diffusion coefficient along x direction (Dx) as a function of slit size at (a) $T = 1200$ K, Pxy = 0 bars when the confining wall is carbon; (b) $T = 1300$ K, Pxy = 0 bars when the confining wall is copper. The diffusion coefficient is calculated by taking derivative of the mean square displacement (MSD).

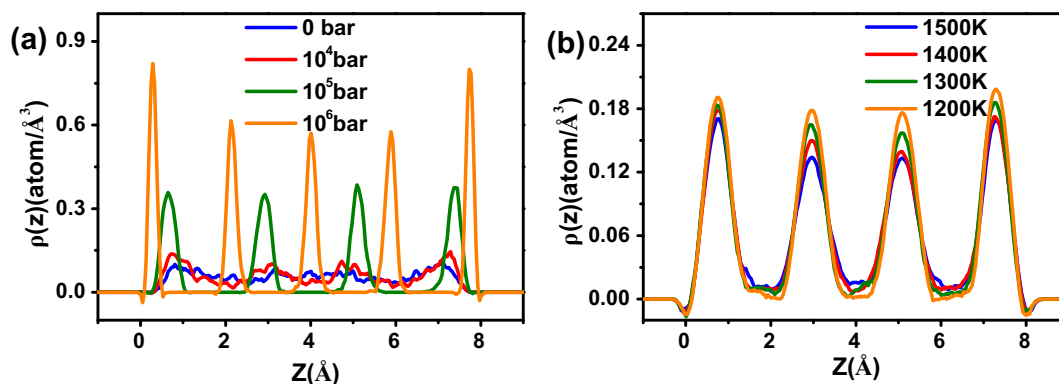


Fig. 6. Density distribution functions (ρ) along z direction for Al atoms confined by carbon wall at (a) $T = 1300$ K, $SS = 12.5$ Å; (b) $P_{xy} = 0$ bars, $SS = 12.5$ Å.

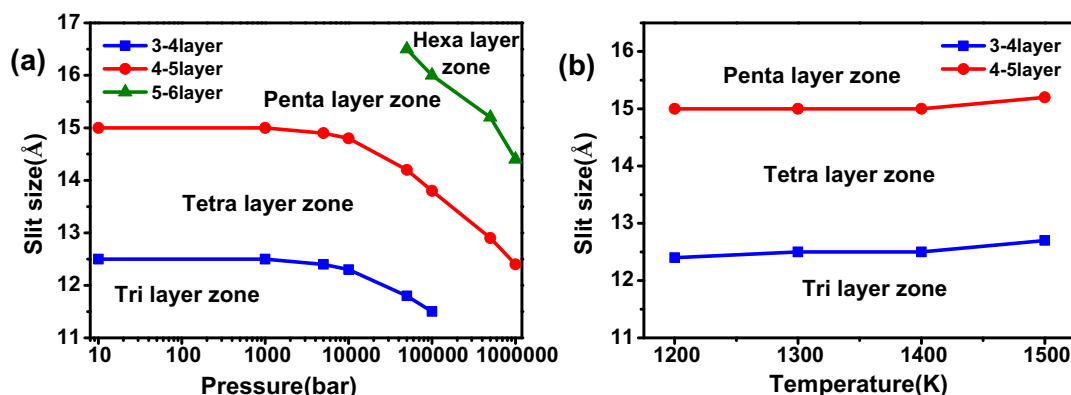


Fig. 7. Phase diagram of the layering transition for Al atoms confined by carbon walls at (a) $T = 1300$ K; (b) $P_{xy} = 0$ bars.

with pressure, indicating that the pressure effect on layering transition points is small. Therefore, the layering transition is mainly SS induced when the pressure is low. While the layering transition line is steep at high pressure zone, suggesting that the layering transition is dominantly high pressure induced. The temperature-SS phase diagram is presented in Fig. 7(b), which is obtained in the absence of the lateral pressure. Compared to the pressure, the temperature has little effect on the layering transition points because the layering transition line is almost parallel to the x axis over the studied temperature range. Changing temperature only cannot cause layering transition when the pressure and SS are fixed. The discrepant effects of pressure and temperature on the layering transition critical points can be attributed to their different effects on the layering as above mentioned.

4. Conclusion

In conclusion, MD simulations are performed to study the layering and its transition of the liquid Al under confinement. The layering, a typical feature of which is the discontinuous distribution of the atomic arrangement and energy, is the result of the size confinement. The layers adjacent to the walls are prone to form denser structure with larger coordination number. Compared to carbon walls, the copper walls are demonstrated to have a stronger confinement effect. We highlight the abnormal changes of the system potential energy and the volume when the layering occurs, indicating its phase transition, which is further validated by the distinctive changes of the coordination number and the diffusion coefficient. The slope difference between pressure-confinement and temperature-confinement diagrams is attributed to the discrepant effects of pressure and temperature on the layering phe-

nomenon. Importantly, under high pressure, the transition is mainly pressure-induced, but under low pressure, it becomes dominantly SS-induced. Our results may give insight into the phase behavior of the liquid metal under the size confinement.

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References

- [1] S.A. Miners, G.A. Rance, A.N. Khlobystov, *Chem. Soc. Rev.* 45 (2016) 4727–4746.
- [2] H. Sha, R. Faller, *Comput. Mater. Sci.* 114 (2016) 160–166.
- [3] Y. Fily, A. Baskaran, M.F. Hagan, *Soft Matter* 10 (2014) 5609–5617.
- [4] C. Paredes-Arroyo, R. Guzmán, *Comput. Mater. Sci.* 131 (2017) 11–20.
- [5] A. Zöttl, H. Stark, *Phys. Rev. Lett.* 112 (2014) 118101.
- [6] S. Zhu, Y. Liu, M. Rafailovich, J. Sokolov, *Nature* 400 (1999) 49.
- [7] T. Prisk, C. Pantalei, H. Kaiser, P. Sokol, *Phys. Rev. Lett.* 109 (2012) 075301.
- [8] C. Cui, L. Gan, M. Neumann, M. Heggen, B. Roldan Cuenya, P. Strasser, *J. Am. Chem. Soc.* 136 (2014) 4813–4816.
- [9] R.C. Remsing, E. Xi, S. Vembanur, S. Sharma, P.G. Debenedetti, S. Garde, A.J. Patel, *Proc. Natl. Acad. Sci. U. S. A.* 112 (2015) 8181–8186.
- [10] Q. Li, B. Wang, Z. Zhao, *Comput. Mater. Sci.* 95 (2014) 121–128.
- [11] C. Alba-Simionesco, B. Coasne, G. Dosseh, G. Dudziak, K. Gubbins, R. Radhakrishnan, M. Sliwinska-Bartkowiak, *J. Phys. Condens. Matter* 18 (2006) R15.
- [12] B. Coasne, J. Czwartos, K.E. Gubbins*, F.R. Hung, M. Sliwinska-Bartkowiak, *Mol. Phys.* 102 (2004) 2149–2163.
- [13] J. Czwartos, B. Coasne, K. Gubbins*, F. Hung, M. Sliwinska-Bartkowiak, *Mol. Phys.* 103 (2005) 3103–3113.

- [14] M. Alcoutlabi, G.B. McKenna, *J. Phys. Condens. Matter* 17 (2005) R461.
- [15] J. Forrest, K. Dalnoki-Veress, J. Dutcher, *Phys. Rev. E* 56 (1997) 5705.
- [16] K. Dalnoki-Veress, J. Forrest, C. Murray, C. Gigault, J. Dutcher, *Phys. Rev. E* 63 (2001) 031801.
- [17] M. Ramanathan, I.S. Michael Kilbey, Q. Ji, J.P. Hill, K. Ariga, *J. Mater. Chem.* 22 (2012) 10389–10405.
- [18] L. Verstraete, J. Greenwood, B.E. Hirsch, S. De Feyter, *ACS Nano* 10 (2016) 10706–10715.
- [19] H. Yabu, T. Higuchi, H. Jinnai, *Soft Matter* 10 (2014) 2919–2931.
- [20] J. Zhou, J. Qin, X. Zhang, C. Shi, E. Liu, J. Li, N. Zhao, C. He, *ACS Nano* 9 (2015) 3837–3848.
- [21] B. Persson, E. Tosatti, *Phys. Rev. B* 50 (1994) 5590.
- [22] M. Milanese, F. Iacobazzi, G. Colangelo, A. de Risi, *Int. J. Heat Mass Transfer* 103 (2016) 564–571.
- [23] T.A. Petach, A. Mehta, R. Marks, B. Johnson, M.F. Toney, D. Goldhaber-Gordon, *ACS Nano* 10 (2016) 4565–4569.
- [24] X. Gong, A. Kozbial, L. Li, *Chem. Sci.* 6 (2015) 3478–3482.
- [25] S. Perkin, *Phys. Chem. Chem. Phys.* 14 (2012) 5052–5062.
- [26] J. Gao, R.S. Ndong, M.B. Shiflett, N.J. Wagner, *ACS Nano* 9 (2015) 3243–3253.
- [27] P. Kowalczyk, A.P. Terzyk, P.A. Gauden, S. Furmaniak, K. Kaneko, T.F. Miller III, *J. Phys. Chem. Lett.* 6 (2015) 3367–3372.
- [28] D. Dietschi, N. Fahl, *Br. Dent. J.* 221 (2016) 765–771.
- [29] N. Giovambattista, P.J. Rossky, P.G. Debenedetti, *Phys. Rev. Lett.* 102 (2009) 050603.
- [30] H. Qiu, X.C. Zeng, W. Guo, *ACS Nano* 9 (2015) 9877–9884.
- [31] D. Alexeev, J. Chen, J.H. Walther, K.P. Giapis, P. Angelikopoulos, P. Koumoutsakos, *Nano Letters* 15 (2015) 5744–5749.
- [32] J.K. Clark II, S.J. Paddison, *Phys. Chem. Chem. Phys.* 16 (2014) 17756–17769.
- [33] M. Shahbabaie, D. Tang, D. Kim, *Comput. Mater. Sci.* 128 (2017) 87–97.
- [34] S. Hanot, S. Lyonard, S. Mossa, *Nanoscale* 8 (2016) 3314–3325.
- [35] W. Li, W. Wang, X. Zheng, Z. Dong, Y. Yan, J. Zhang, *Comput. Mater. Sci.* 136 (2017) 60–66.
- [36] M. Sattig, S. Reutter, F. Fajara, M. Werner, G. Buntkowsky, M. Vogel, *Phys. Chem. Chem. Phys.* 16 (2014) 19229–19240.
- [37] W. Saslow, G. Agnolet, C. Campbell, B. Clements, E. Krotscheck, *Phys. Rev. B* 54 (1996) 6532.
- [38] Y. He, H. Li, Y. Jiang, X. Li, X. Bian, *Sci. Rep.* 4 (2014) 3635.
- [39] Y. He, X. Li, H. Li, Y. Jiang, X. Bian, *Nanoscale* 6 (2014) 4217–4224.
- [40] W. Wu, L. Zhang, S. Liu, H. Ren, X. Zhou, H. Li, *J. Am. Chem. Soc.* 138 (2016) 2815.
- [41] X. Zhou, S. Liu, L. Wang, Y. Li, W. Wu, Y. Duan, H. Li, *Nanoscale* (2016) 8.
- [42] T. Li, W. Wu, H. Li, *Phys. Chem. Chem. Phys.* 18 (2016) 27500–27506.
- [43] Y.-R. Duan, T. Li, W.-K. Wu, J. Li, X.-Y. Zhou, S.-D. Liu, H. Li, *Chin. Phys. B* 26 (2017) 036401.
- [44] X.-Q. Zhang, R.A. van Santen, E.J. Hensen, *ACS Catal.* 5 (2014) 596–601.
- [45] X.-Q. Zhang, E. Iype, S.V. Nedea, A.P. Jansen, B.M. Szyja, E.J. Hensen, R.A. van Santen, *J. Phys. Chem. C* 118 (2014) 6882–6886.
- [46] C.R. Dandekar, Y.C. Shin, *Composites, Part A* 42 (2011) 355–363.